MIL-T-9166B
11 January 1968
SUPERSEDING
MIL-T-9166A(ASG)
17 August 1954

MILITARY SPECIFICATION

TIE DOWN, CARGO, AIRCRAFT, CABLE NET, 10,000 LB CAPACITY, TYPE A-2

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a cargo net tie down of cable construction, designated type A-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-W-1511	Wire Rope, Steel (Carbon) Flexible, Preformed
MIL-T-7181	Tie Down, Cargo, Aircraft, Type A-1A
MIL-A-8625	Anodic Coating, For Aluminum And Aluminum Alloys

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property

FSC 1670

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MIL-STD-143 Specifications And Standards Order Of Precedence For The Selection Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Materials. Parts fabricated from metals that are not corrosion resistant shall be suitably protected to resist corrosion in extremes of atmospheric conditions likely to occur during normal service life.

3.4 Design and construction. The tie down shall be so designed and constructed that no parts will work loose in service. The tie down shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Cable. The steel cable shall conform to MIL-W-1511. Cable ends shall be covered or otherwise protected to prevent fraying and preclude protruding strands.

3.4.2 Strength. The breaking strength of each primary member of the tie down shall be 2,000 pounds minimum. The breaking strength of each secondary member of the net shall be 1,000 pounds minimum. Primary members shall be those members intersecting on 20-inch centers to which attachment fittings are anchored.

3.4.3 Mesh. The mesh shall be a diamond configuration, each side having a length of 7 ±1/2 inches.

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3.4.3.1 Where portions of the cable intersect in forming the mesh, swage fittings shall be utilized. The strands of the cable shall not be interlaced in any manner.

3.4.4 Attachment fittings. Rings shall be provided on two adjacent sides of the tie down. Hooks shall be provided on the remaining two adjacent sides.

3.4.4.1 The attachment fittings shall be spaced on 20-inch centers along the perimeter of the tie down starting at one corner.

3.4.4.2 The nominal strength of the rings and hooks shall be 1,250 pounds minimum.

3.4.4.3 The hooks shall be guarded, and the dimensions shall be suitable for attachment to floor rings having a bar stock diameter of 3/4 inch, and to the rings of the adjacent tie down.

3.4.4.4 The ring dimensions shall be suitable for attachment of a type A-1A cargo tie down in accordance with MIL-T-7181.

3.4.4.5 The attachment fittings shall not be secured to the tie down by the use of any type of fabric.

3.4.5 Stowage. The tie down shall be designed to fold or coil into a compact unit in order that the space occupied for stowage shall not exceed 30 inches in diameter and 12 inches in depth. Stowage of the tie down shall be possible for one person to accomplish in approximately 2 minutes.

3.5 Maintainability. The tie down shall be so constructed that a minimum of maintenance is required and that adjustments and repairs can be easily made by personnel of operating units and overhaul bases.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

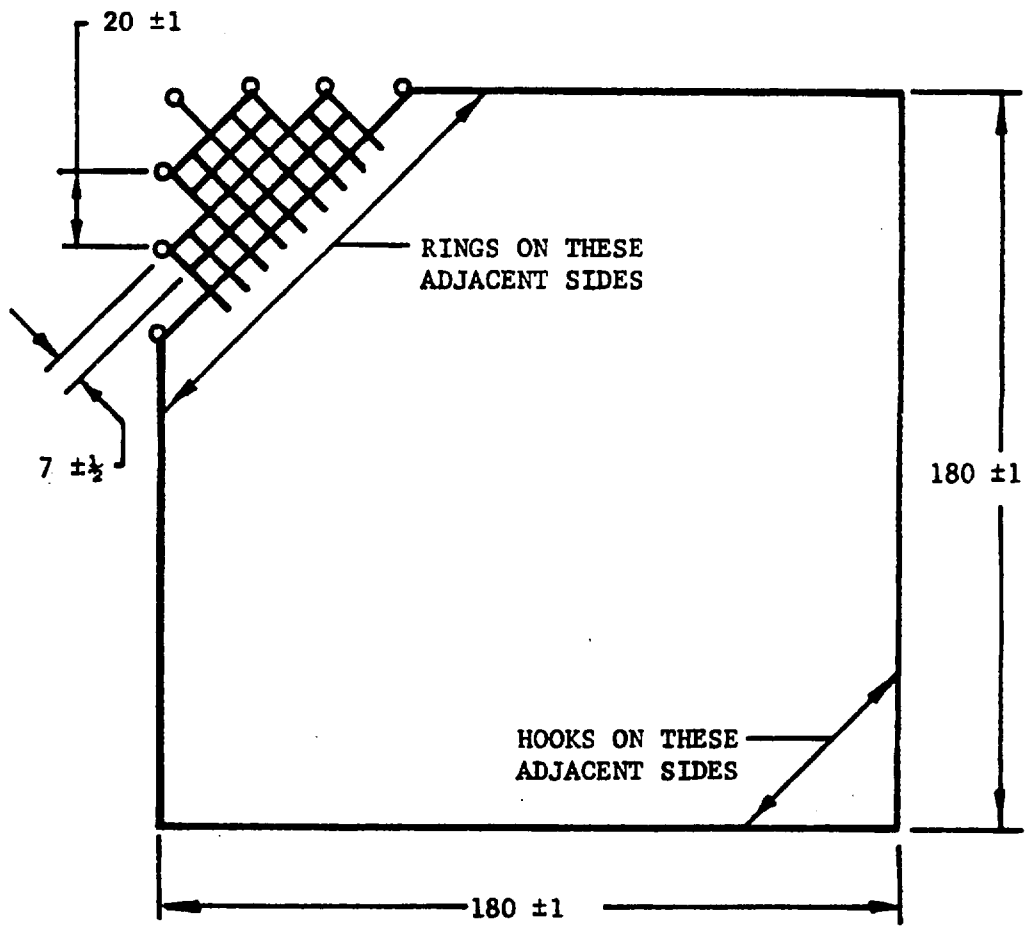
3.7 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the tie down, they shall be held or limited accordingly. The tie down shall conform to Figure 1.

3.8 Weight. The weight of the tie down shall not exceed 42 pounds.

3.9 Finishes and protective coatings.

3.9.1 The tie down shall be thoroughly cleaned, and metal chips and other foreign material shall be removed during and after final assembly.

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DIMENSIONS IN INCHES.

FIGURE 1

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3.9.2 All steel parts, with the exception of the cable, shall be cadmium plated in accordance with QQ-P-416, type I, class I.

3.9.3 All aluminum-alloy parts shall be anodized in accordance with MIL-A-8625.

3.10 Identification of product. The assembly shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship. The tie down, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from defects, burrs, and sharp edges; accuracy of dimensions and marking of parts and assemblies; thoroughness of swaging; alinement of parts; et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of tie downs shall be classified as follows:

a. Preproduction tests (see 4.3)

b. Acceptance tests (see 4.4)

4.3 Preproduction tests.

4.3.1 Sampling instructions. The samples specified in 4.3.2.1, 4.3.2.2, and 4.3.2.3, as specified in the contract, will be tested for design approval by the procuring activity or, when so specified in the contract, at the contractor's plant under the supervision of a government representative.

4.3.2 Tests. The preproduction tests of tie downs shall consist of all the tests specified under 4.4 and, in addition, the following tests.

4.3.2.1 Five swaged fittings of the type used for the tie down shall be tested as follows: Two pieces of cable approximately 7 inches long, from the lot to be used in the construction of the tie downs, shall be placed along side of each other. A swage fitting from the lot to be used in assembling the tie downs shall be swaged to 2 of the adjacent ends of the cables allowing at least 1/2 inch overhang of the cables. The process and equipment

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utilized in fabricating the tie downs shall be used in preparing these specimens. The 2 remaining free ends of the cables shall be so rotated that they are 180 degrees apart. The free ends shall then be placed in opposing ends of a testing machine. This assembly will then be pull tested. Slippage shall not occur below 900 pounds.

4.3.2.2 Two ring attachment assemblies, with approximately 7 inches of free cable from the swaging, shall be pull tested. No permanent deformation of the ring or fracture of any part shall occur below 1,250 pounds.

4.3.2.3 Two hook attachment assemblies, with approximately 7 inches of free cable from the swaging, shall be pull tested. No permanent deformation of the hook or fracture of any part shall occur below 1,250 pounds.

4.3.2.4 After the tests specified in 4.3.2.2 and 4.3.2.3, the ring and hook assemblies shall be further tested to insure a minimum failure load of 2,000 pounds.

4.4 Acceptance tests. The acceptance tests shall consist of individual tests and sampling tests.

4.4.1 Individual tests. Each tie down shall be subjected to the following test.

4.4.1.1 Examination of product. Each tie down shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2 Sampling tests. One hundred sample swaged fittings shall be fabricated and tested in accordance with 4.3.2.1 at random during the course of fabrication of each lot of 500 complete tie down nets.

4.4.2.1 The samples shall be subjected to the test specified in 4.3.2.1.

4.4.3 Rejection and retest. When sampling tests are specified on a number of tie downs that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, acceptance of all tie downs shall be withheld until the extent and cause of failure are determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of the tie downs is contingent upon the inspector's decision regarding the overall conformance of the tie downs to specification requirements. If investigation indicates that the defects may exist on tie downs previously accepted, full particulars concerning the defects found, including recommendations for correction, shall be furnished to the procuring activity.

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4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Tie down assemblies shall be preserved and packaged in accordance with MIL-P-116, method III, one each in a weather-resistant, snug fitting unit container conforming to Style RSC of PPP-B-636. Each box shall be closed and waterproofed in accordance with the appendix to PPP-B-636.

5.1.2 Level C. Preservation and packaging shall afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use or controlled humidity storage. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Tie down assemblies preserved and packaged as specified in 5.1.1 shall be packed in overseas type shipping containers conforming to PPP-B-601. As far as practicable, shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. The container shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.2 Level B. Each tie down assembly, preserved and packaged as specified in 5.1.1, need not be overpacked. The inside of each fiberboard box shall be fitted with a taped liner conforming to class domestic, variety DW, grade 275 of PPP-B-636.

5.2.3 Level C. Packing shall afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type A-2 tie down covered by this specification is intended for use in cargo aircraft for in-flight restraint of miscellaneous cargo.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging and level of packing (see 5.1 and 5.2).

c. Preproduction sample. It is expected that the contract or purchase order will specify that samples specified in 4.3.1 will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - AS
Army - GL
Air Force - 82

Review Activities:
Navy - AS
Army - AV
Air Force - 82

User:
Army - SM

Preparing Activity:
Air Force - 82

Project No. 1670-0159

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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